chance to react. Now, sixth-gen fighters like the F-47 and J-36 promise another leap forward, incorporating AI, drone wingmen and hypersonic weaponry. The implications are stark: by the time India's AMCA becomes operational, its technological edge may already be blunted against more advanced adversaries.

Speaking at the India Today Conclave on March 8, Air Chief Marshal A.P. Singh highlighted the competitive pressure from China's advancements and the necessity to expedite India's own fighter jet development, particularly the AMCA. "India needs to catch up with the latest technology and accelerate its fifth-gen fighter jet programme at a time when China has showcased its sixth-gen combat aircraft," he said. The Chief of the Air Staff also talked of stepping up production, emphasising the "need to add 35-40 fighter jets every year to fill the shortage".

At a time the world is focusing on air dominance, the IAF is struggling to fill the numbers of its sanctioned fleet. It is left with barely 30 squadrons, against the sanctioned strength of 42, with at least eight more set to retire over the next decade. Each squadron typically comprises 18-20 jets, which means the IAF presently has a fleet of around 540-600 fighters. To address the dwindling numbers and modernise the air force, a previous defence secretary-led committee had handed over a detailed roadmap to Union defence minister Rajnath Singh on March 3. The immediate goal is to reach the sanctioned strength and then scale it up to 60 squadrons by 2047. The AMCA is a crucial part of this vision (see *Roadmap 2047: IAF's Expansion Plan*).

THE FIFTH-GEN CHALLENGE

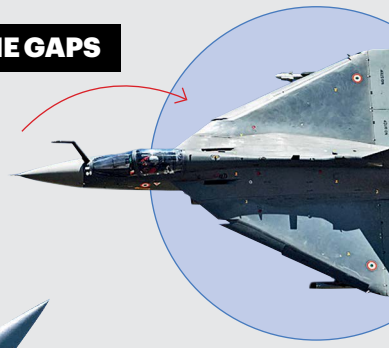
The AMCA project, so far, has been marred by delays. It was in 2009 that the Union government commissioned a feasibility study, but it was only in March 2024 that the project got the Union cabinet's go-ahead, with an initial allocation of Rs 15,000 crore for the development of five prototypes. A 10-year development roadmap has been finalised for their rollout for rigorous flight tests. Even the IAF's modernisation roadmap envisions inducting around seven AMCA squadrons beginning in 2035.

While approving the project, the Cabinet Committee on Security (CCS) made it clear that any cost overruns and delays should be directly addressed to it instead of the defence ministry. While some experts highlight the high cost, the AMCA remains significantly more affordable than other stealth jet programmes. The NGAD, which includes the F-47 and its next-gen engine, is projected to exceed $40 billion (Rs 3.4 lakh crore). Similarly, the UK-Italy-Japan Global Combat Air Programme (GCAP) is expected to cost $43.2 billion (Rs 3.7 lakh crore). In contrast, India's combined Light Combat Aircraft (LCA) Mk2, Twin Engine Deck-Based Fighter (TEDBF) and AMCA programmes are estimated at $4.6 billion (Rs 40,000 crore).

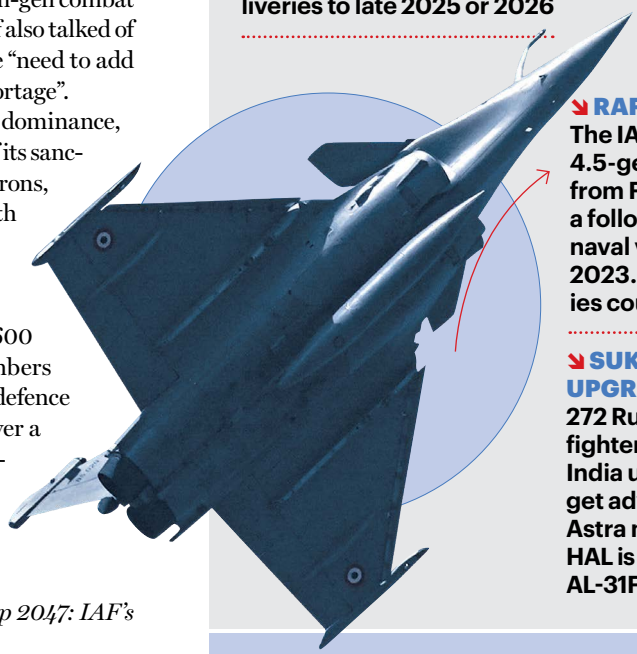
ROADMAP 2047
IAF'S EXPANSION PLAN

With just 31 squadrons against a sanctioned 42, the IAF faces a dire shortfall. Ageing MiG-21s, Jaguars and Mirage 2000s will phase out by 2030, outpacing new inductions. The goal: 42 squadrons ASAP, scaling to 60 (i.e. 1,080-1,200 fighter jets) by 2047

2025-30: PLUGGING THE GAPS



TEJAS MARK-1A: With more than 40 improvements over the Mark-1 variant, 180 of these 4.5-gen Light Combat Aircraft to replace MiG-21s (9-10 squadrons). GE F404 engine supply delays push deliveries to late 2025 or 2026



RAFALE EXPANSION The IAF has 36 of these 4.5-gen fighters procured from France. Reports of a follow-on order for 26 naval variants surfaced in 2023. If greenlit, deliveries could begin by 2028



SUKHOI SU-30MKI UPGRADES: In the fleet of 272 Russian-origin 4.5-gen fighters manufactured in India under licence, 84 to get advanced AESA radar, Astra missiles from 2026. HAL is delivering new AL-31FP engines too

GENERATIONAL LEAP

The evolution of fighter jets

The first three generations of fighter jets evolved from the pioneering 1st-gen Me 262 (Germany) during the WW-II and the Korean War's F-86 (US) and MiG-15 (Soviet Union) to supersonic MiG-21 and F-4 Phantom (2nd/ 3rd gen) engaged in the Vietnam war—each leap bringing faster speeds, advanced missiles and multirole capabilities

4th-gen fighters like Russia's Su-27 and the US's F-16 dominated late 20th-century conflicts with